Deploy Prometheus and Grafana on Kubernetes using Terraform & Helm

This document provides complete step-by-step practical implementation for deploying Prometheus and Grafana on a Kubernetes cluster using Terraform and Helm.

1. Objective

To automate deployment of Prometheus and Grafana monitoring stack on Kubernetes using Infrastructure as Code (IaC) tools such as Terraform and Helm.

2. Technologies Used

• Kubernetes
• Minikube
• Terraform
• Helm
• Prometheus
• Grafana
• Docker Desktop
• kubectl
• VS Code

3. Software Installation

Install the following software:
1. Docker Desktop
2. Minikube
3. kubectl
4. Terraform
5. Helm
6. VS Code

4. Start Kubernetes Cluster

Open Command Prompt and run:

minikube start

Verify cluster:

kubectl get nodes

5. Create Project Folder

Create folder:

monitoring-project

Create files:
• provider.tf
• main.tf
• variables.tf
• outputs.tf

6. Terraform Provider Configuration

provider.tf

```
terraform {
  required_providers {

    kubernetes = {
      source  = "hashicorp/kubernetes"
      version = "2.27.0"
    }

    helm = {
      source  = "hashicorp/helm"
      version = "2.13.2"
    }
  }
}

provider "kubernetes" {
  config_path = "C:/Users/SAMRUDDHI/.kube/config"
}

provider "helm" {
  kubernetes {
    config_path = "C:/Users/SAMRUDDHI/.kube/config"
  }
}
```

7. Terraform Main Configuration

main.tf

```
resource "kubernetes_namespace" "monitoring" {

  metadata {
    name = "monitoring"
  }
}
```

```
resource "helm_release" "prometheus" {

  name       = "prometheus"

  repository = "https://prometheus-community.github.io/helm-charts"

  chart      = "prometheus"

  namespace  = kubernetes_namespace.monitoring.metadata[0].name
}

resource "helm_release" "grafana" {

  name       = "grafana"

  repository = "https://grafana.github.io/helm-charts"

  chart      = "grafana"

  namespace  = kubernetes_namespace.monitoring.metadata[0].name

  values = [
    <<EOF
adminPassword: admin123
EOF
  ]
}
```

8. Terraform Commands

Initialize Terraform:

```
terraform init
```

Validate:

```
terraform validate
```

Plan:

```
terraform plan
```

Deploy:

terraform apply

9. Verify Deployment

Check Pods:

kubectl get pods -n monitoring

Check Services:

kubectl get svc -n monitoring

Check Helm Releases:

helm list -n monitoring

10. Access Prometheus

Run:

kubectl port-forward svc/prometheus-server 9090:80 -n monitoring

Open Browser:
http://localhost:9090

11. Access Grafana

Run:

kubectl port-forward svc/grafana 3000:80 -n monitoring

Open Browser:
http://localhost:3000

Username: admin
Password: admin123

12. Configure Grafana

1. Open Grafana
2. Go to Connections → Data Sources
3. Add Prometheus
4. Enter URL:

http://prometheus-server.monitoring.svc.cluster.local

5. Save & Test

13. Import Dashboard

Use Dashboard ID:

1860

This dashboard displays Kubernetes monitoring metrics.

14. Screenshots Section

Add the following screenshots in your GitHub repository and report:

1. kubectl get pods -n monitoring
2. Prometheus Dashboard
3. Grafana Dashboard

15. Benefits

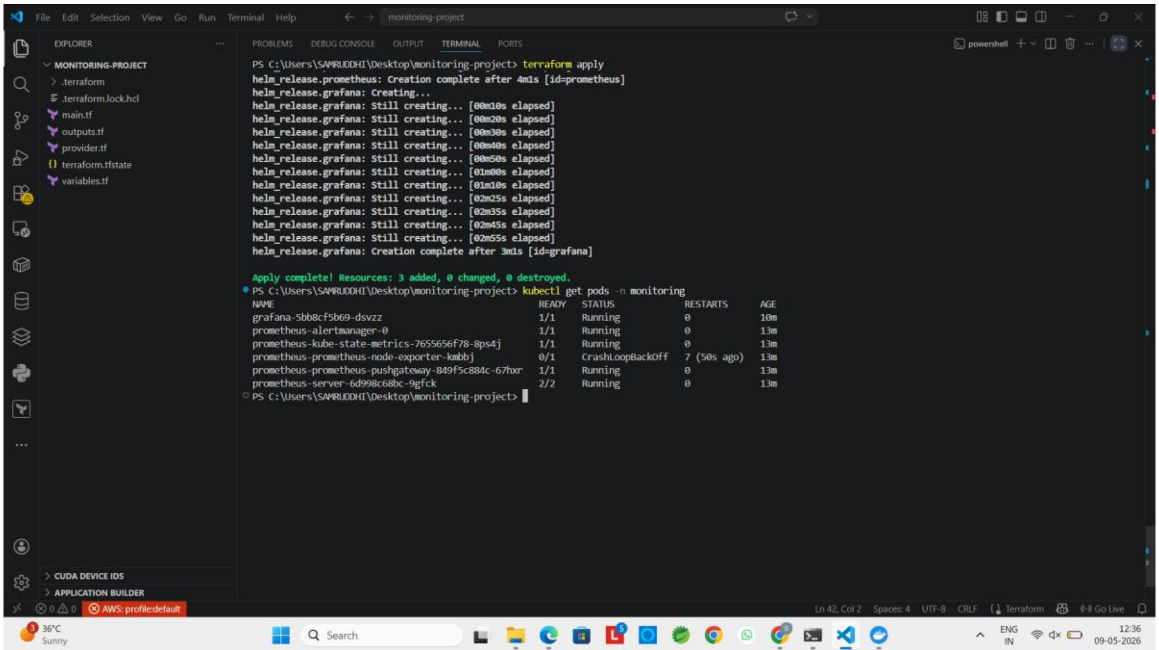
Benefits of Terraform:
• Infrastructure automation
• Consistency
• Reusability

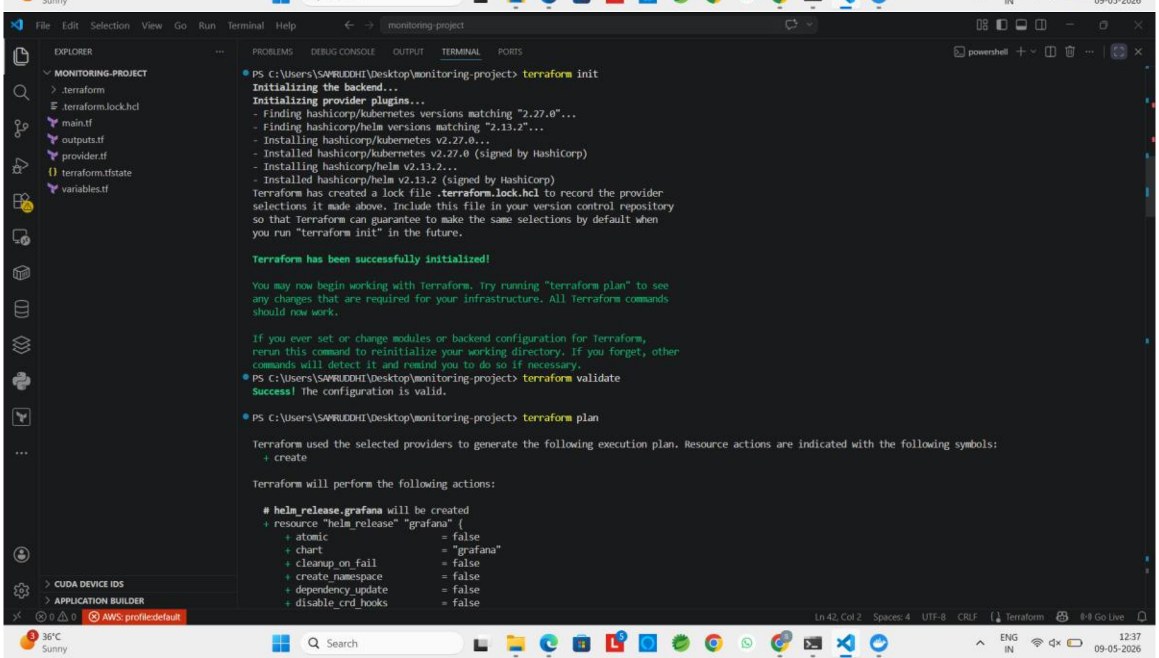
Benefits of Helm:
• Easy Kubernetes deployment
• Package management
• Simplified upgrades

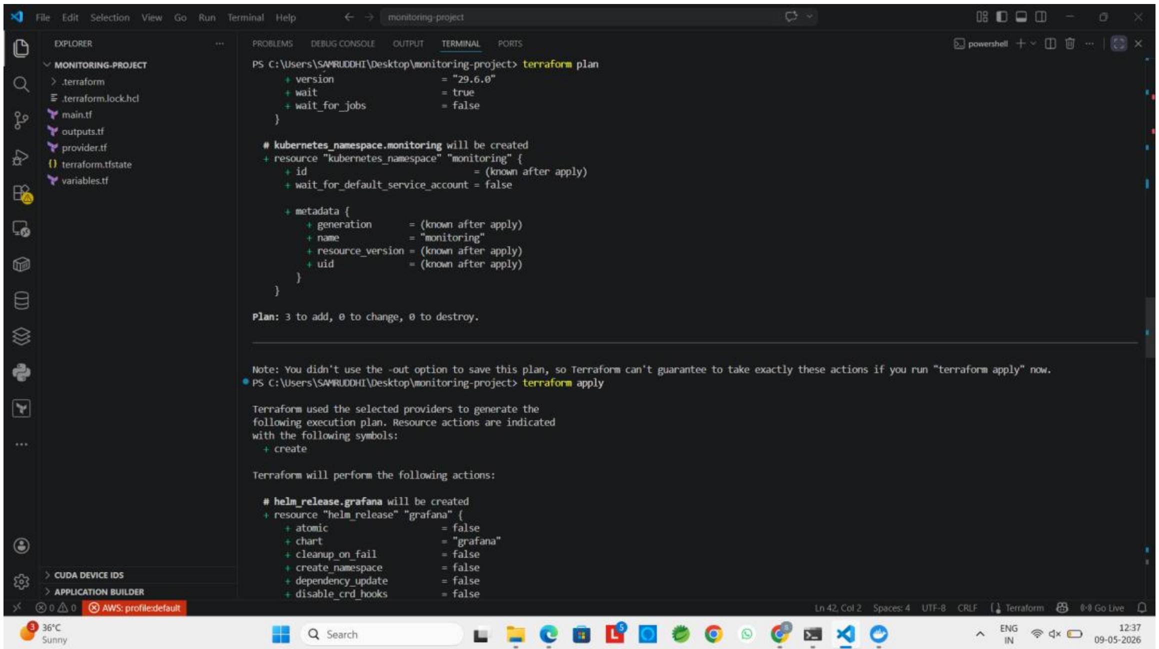
16. Conclusion

This project successfully demonstrates deployment of Prometheus and Grafana on Kubernetes using Terraform and Helm. The monitoring stack provides real-time visualization and monitoring of Kubernetes infrastructure.

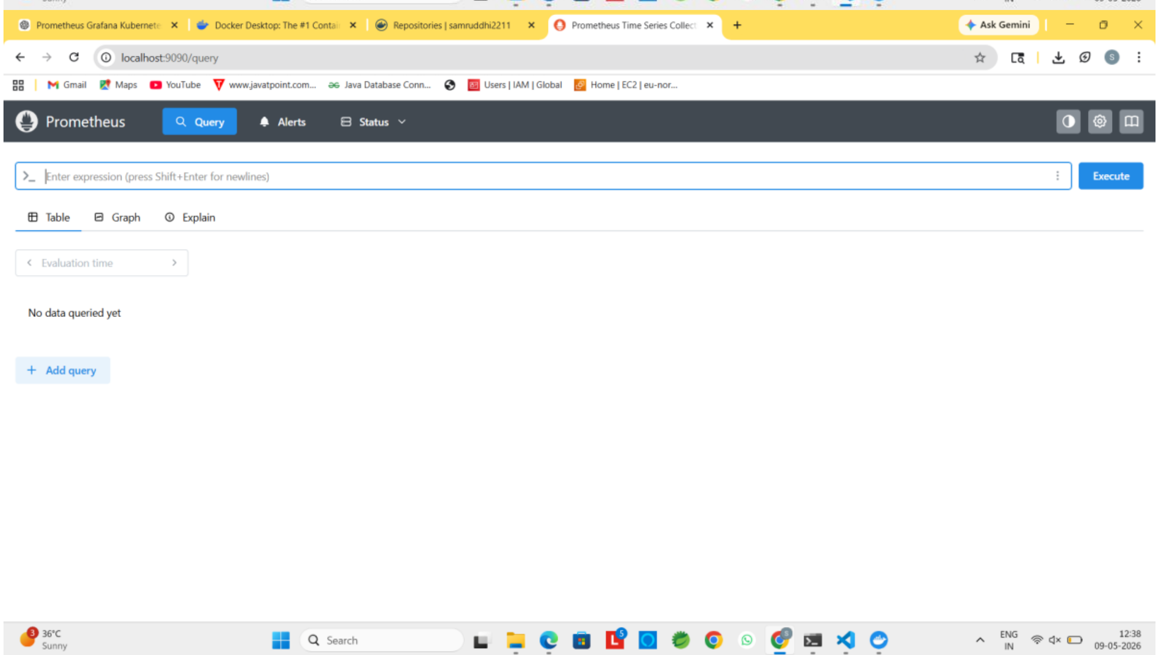
NOTE: Insert your screenshots at the appropriate sections before final submission.



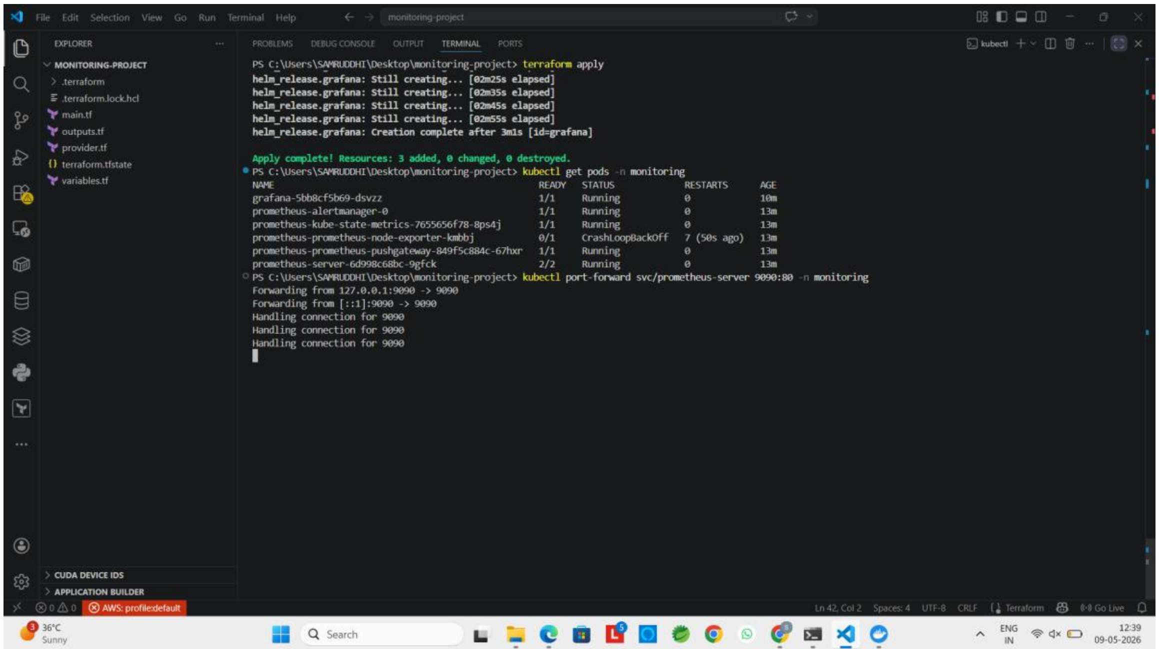


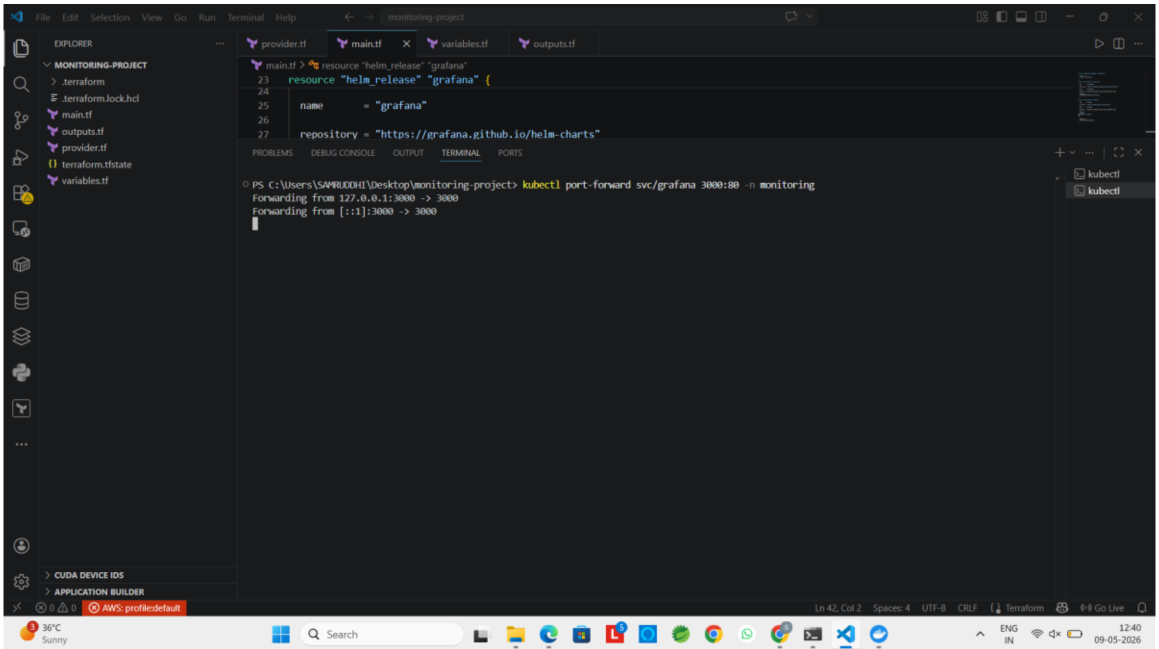


File Edit Selection View Go Run Terminal Help
monitoring-project
EXPLORER
MONITORING-PROJECT
.terraform
.terraform.lock.hcl
main.tf
outputs.tf
provider.tf
terraform.tfstate
variables.tf
CUDA DEVICE IDS
APPLICATION BUILDER
PROBLEMS DEBUG CONSOLE OUTPUT TERMINAL PORTS
powershell
PS C:\Users\SAMRUDDHI\Desktop\monitoring-project> terraform plan
+ version = "29.6.0"
+ wait = true
+ wait_for_jobs = false
}
# kubernetes_namespace.monitoring will be created
+ resource "kubernetes_namespace" "monitoring" {
+ id = (known after apply)
+ wait_for_default_service_account = false
+ metadata {
+ generation = (known after apply)
+ name = "monitoring"
+ resource_version = (known after apply)
+ uid = (known after apply)
}
}
Plan: 3 to add, 0 to change, 0 to destroy.
Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.
PS C:\Users\SAMRUDDHI\Desktop\monitoring-project> terraform apply
Terraform used the selected providers to generate the
following execution plan. Resource actions are indicated
with the following symbols:
+ create
Terraform will perform the following actions:
# helm_release.grafana will be created
+ resource "helm_release" "grafana" {
+ atomic = false
+ chart = "grafana"
+ cleanup_on_fail = false
+ create_namespace = false
+ dependency_update = false
+ disable_crd_hooks = false
AWS: profile:default
Ln 42, Col 2 Spaces: 4 UTF-8 CRLF Terraform Go Live
36°C
Sunny
Search
ENG
IN
12:37
09-05-2026

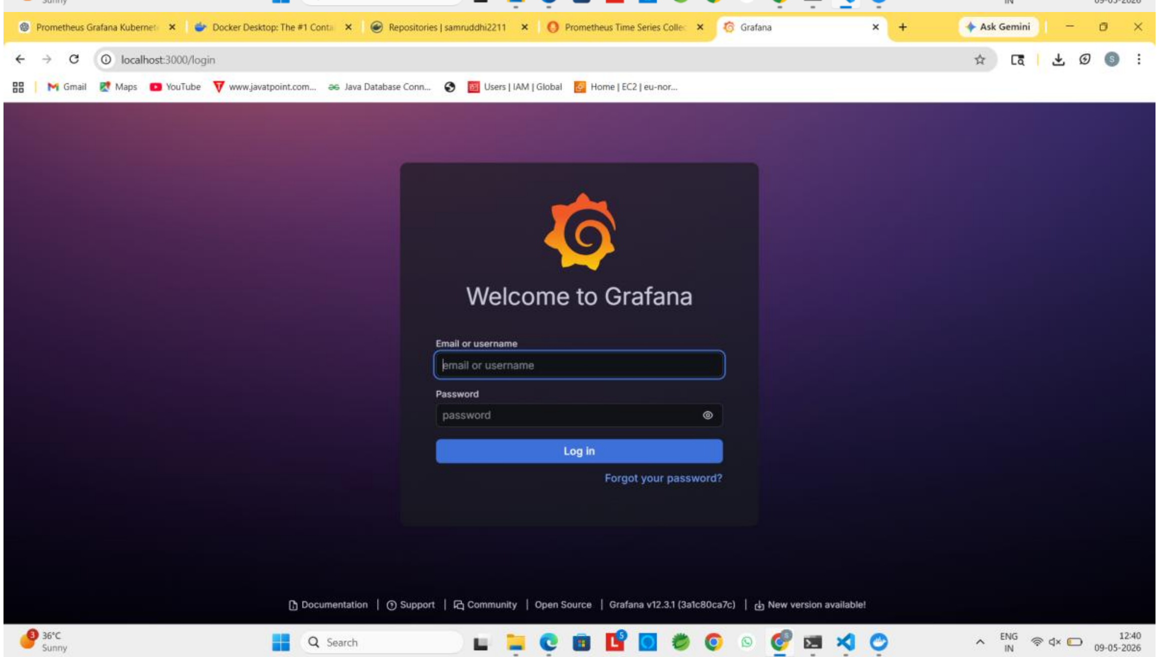


Prometheus Grafana Kubernete
Docker Desktop: The #1 Conta
Repositories | samruddhi2211
Prometheus Time Series Collect
Ask Gemini
localhost:9090/query
Gmail
Maps
YouTube
www.javatpoint.com...
Java Database Conn...
Users | IAM | Global
Home | EC2 | eu-nor...
Prometheus
Query
Alerts
Status
Enter expression (press Shift+Enter for newlines)
Execute
Table
Graph
Explain
Evaluation time
No data queried yet
Add query
36°C
Sunny
Search
ENG
IN
12:38
09-05-2026

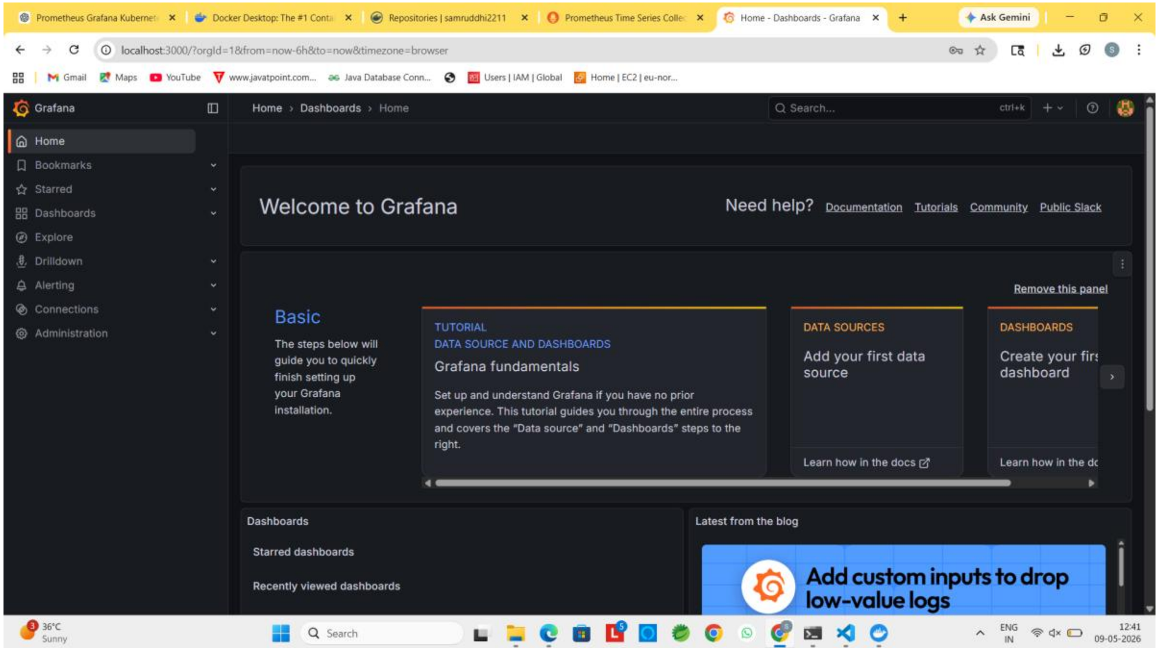
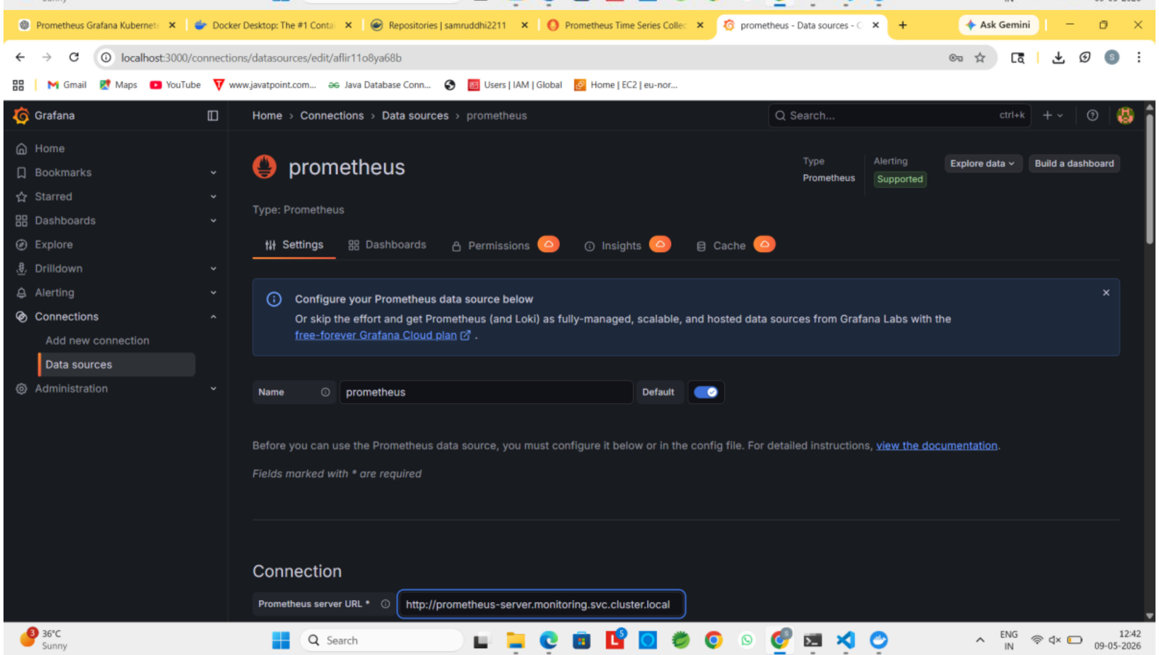

File Edit Selection View Go Run Terminal Help
monitoring-project
EXPLORER
MONITORING-PROJECT
.terraform
.terraform.lock.hcl
main.tf
outputs.tf
provider.tf
terraform.tfstate
variables.tf
PROBLEMS DEBUG CONSOLE OUTPUT TERMINAL PORTS
PS C:\Users\SAMRUDDHI\Desktop\monitoring-project> terraform apply
helm_release.grafana: Still creating... [02m25s elapsed]
helm_release.grafana: Still creating... [02m35s elapsed]
helm_release.grafana: Still creating... [02m45s elapsed]
helm_release.grafana: Still creating... [02m55s elapsed]
helm_release.grafana: Creation complete after 3m1s [id=grafana]
Apply complete! Resources: 3 added, 0 changed, 0 destroyed.
PS C:\Users\SAMRUDDHI\Desktop\monitoring-project> kubectl get pods -n monitoring
NAME READY STATUS RESTARTS AGE
grafana-5bb8cf5b69-dsvzz 1/1 Running 0 10m
prometheus-alertmanager-0 1/1 Running 0 13m
prometheus-kube-state-metrics-7655656f78-8ps4j 1/1 Running 0 13m
prometheus-prometheus-node-exporter-kmbbj 0/1 CrashLoopBackOff 7 (50s ago) 13m
prometheus-prometheus-pushgateway-849f5c884c-67hxr 1/1 Running 0 13m
prometheus-server-6d998c68bc-9gfck 2/2 Running 0 13m
PS C:\Users\SAMRUDDHI\Desktop\monitoring-project> kubectl port-forward svc/prometheus-server 9090:80 -n monitoring
Forwarding from 127.0.0.1:9090 -> 9090
Forwarding from [::1]:9090 -> 9090
Handling connection for 9090
Handling connection for 9090
Handling connection for 9090
CUDA DEVICE IDS
APPLICATION BUILDER
AWS: profile:default
Ln 42, Col 2 Spaces: 4 UTF-8 CRLF Terraform Go Live
36°C
Sunny
Search
ENG
IN
12:39
09-05-2026

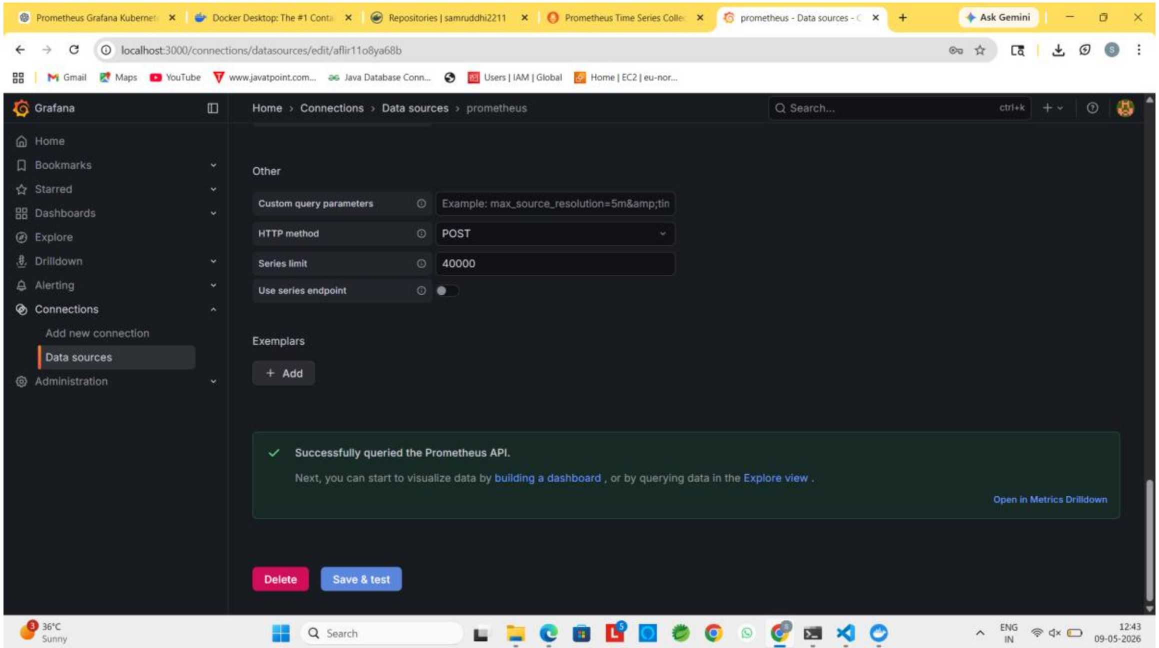


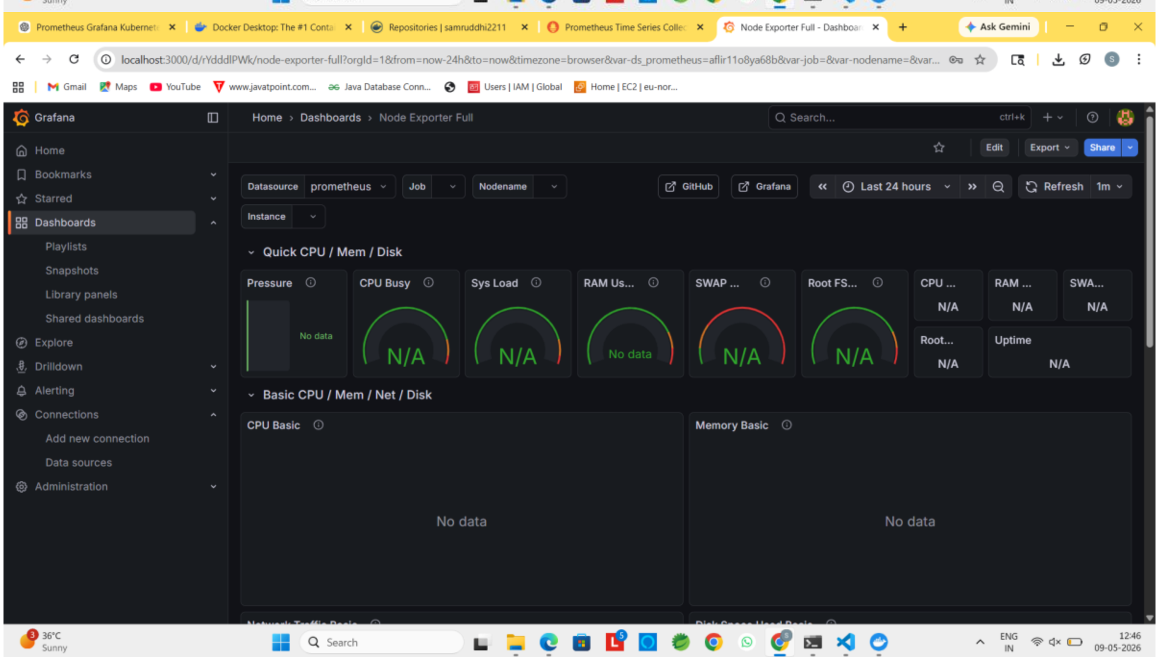
EXPLORER
MONITORING-PROJECT
.terraform
.terraform.lock.hcl
main.tf
outputs.tf
provider.tf
terraform.tfstate
variables.tf
provider.tf
main.tf
variables.tf
outputs.tf
resource "helm_release" "grafana" {
name = "grafana"
repository = "https://grafana.github.io/helm-charts"
PROBLEMS
DEBUG CONSOLE
OUTPUT
TERMINAL
PORTS
PS C:\Users\SAMRUDDHI\Desktop\monitoring-project> kubectl port-forward svc/grafana 3000:80 -n monitoring
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
kubectl
CUDA DEVICE IDS
APPLICATION BUILDER
AWS: profile:default
Ln 42, Col 2
Spaces: 4
UTF-8
CRLF
Terraform
Go Live
36°C
Sunny
Search
12:40
09-05-2026

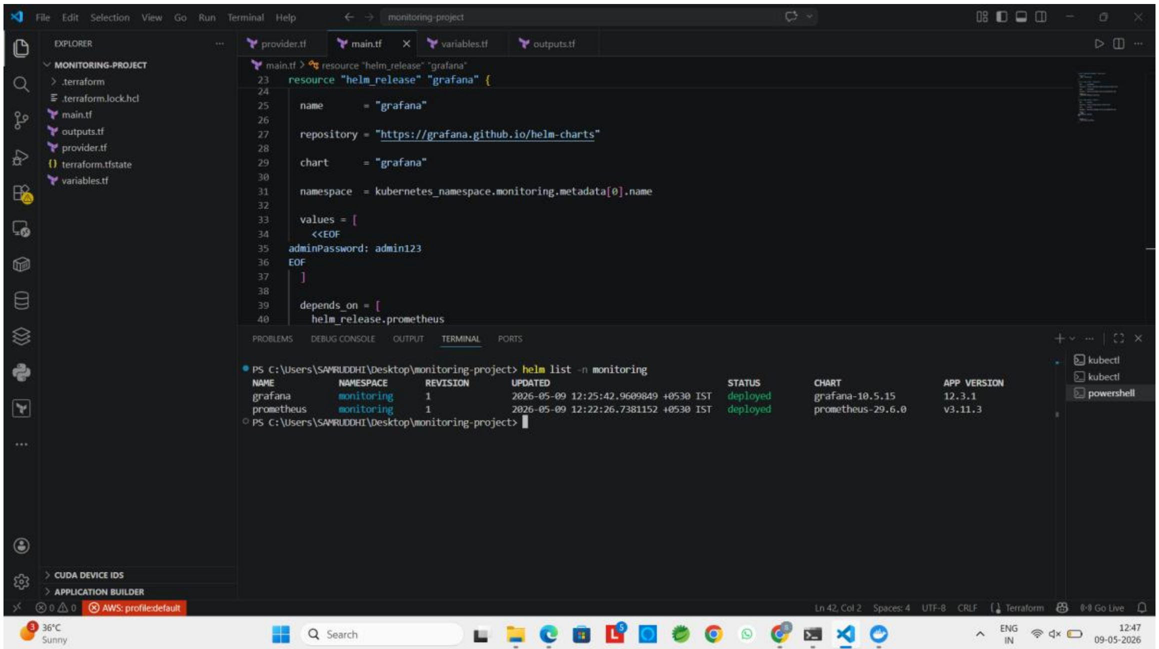


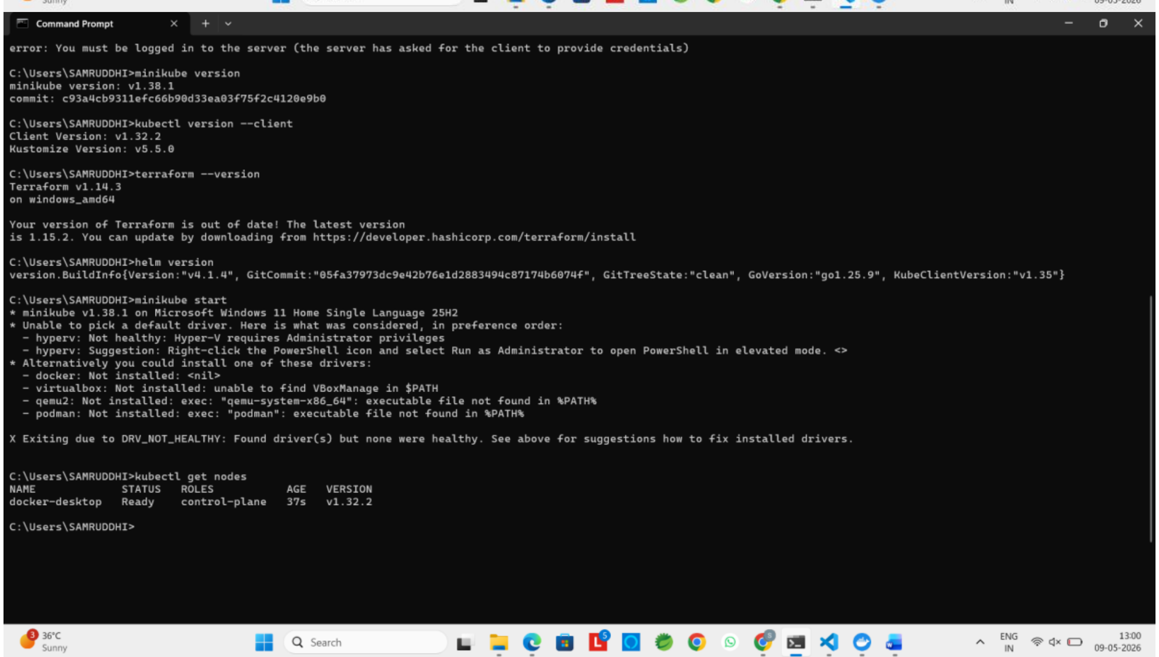
Grafana
localhost:3000/login
Welcome to Grafana
Email or username
email or username
Password
password
Log in
Forgot your password?
Documentation | Support | Community | Open Source | Grafana v12.3.1 (3a1c80ca7c) | New version available!
36°C
Sunny
Search
12:40
09-05-2026

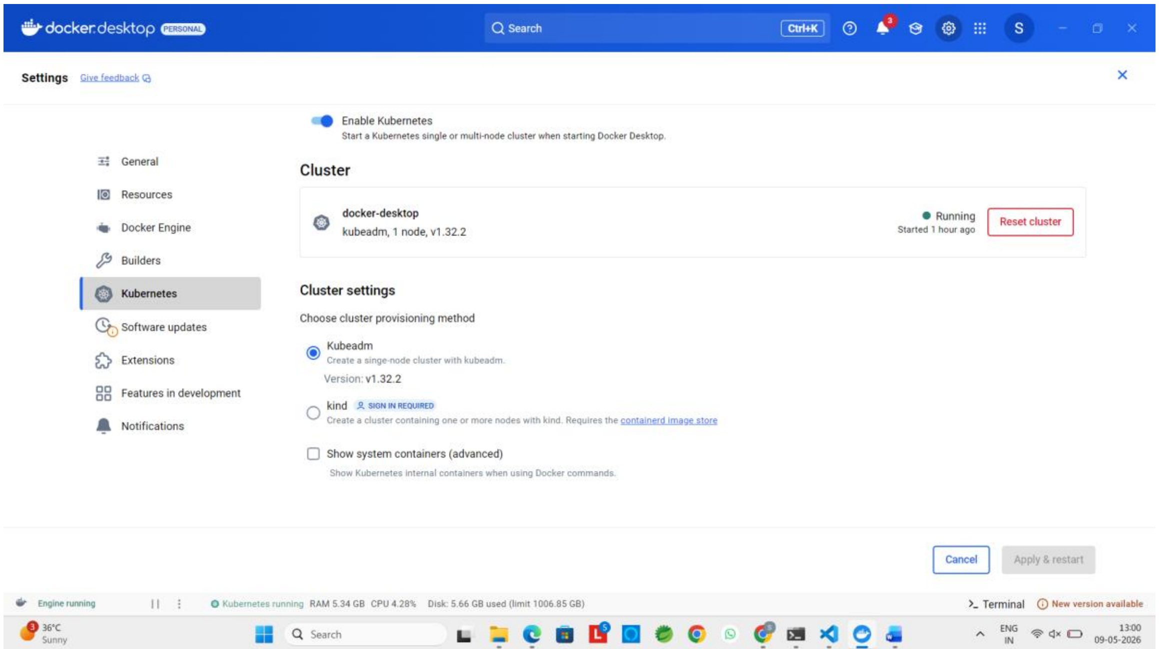

localhost:3000/?orgId=1&from=now-6h&to=now&timezone=browser
Grafana
Home › Dashboards › Home
Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
Alerting
Connections
Administration
Welcome to Grafana
Need help? Documentation Tutorials Community Public Slack
Remove this panel
Basic
The steps below will guide you to quickly finish setting up your Grafana installation.
TUTORIAL
DATA SOURCE AND DASHBOARDS
Grafana fundamentals
Set up and understand Grafana if you have no prior experience. This tutorial guides you through the entire process and covers the "Data source" and "Dashboards" steps to the right.
DATA SOURCES
Add your first data source
Learn how in the docs
DASHBOARDS
Create your firs dashboard
Dashboards
Starred dashboards
Recently viewed dashboards
Latest from the blog
Add custom inputs to drop low-value logs

localhost:3000/connections/datasources/edit/aflr11o8ya68b
Grafana
Home › Connections › Data sources › prometheus
Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
Alerting
Connections
Add new connection
Data sources
Administration
prometheus
Type: Prometheus
Type Prometheus
Alerting Supported
Explore data
Build a dashboard
Settings
Dashboards
Permissions
Insights
Cache
Configure your Prometheus data source below
Or skip the effort and get Prometheus (and Loki) as fully-managed, scalable, and hosted data sources from Grafana Labs with the free-forever Grafana Cloud plan .
Name
prometheus
Default
Before you can use the Prometheus data source, you must configure it below or in the config file. For detailed instructions, view the documentation.
Fields marked with * are required
Connection
Prometheus server URL *
http://prometheus-server.monitoring.svc.cluster.local

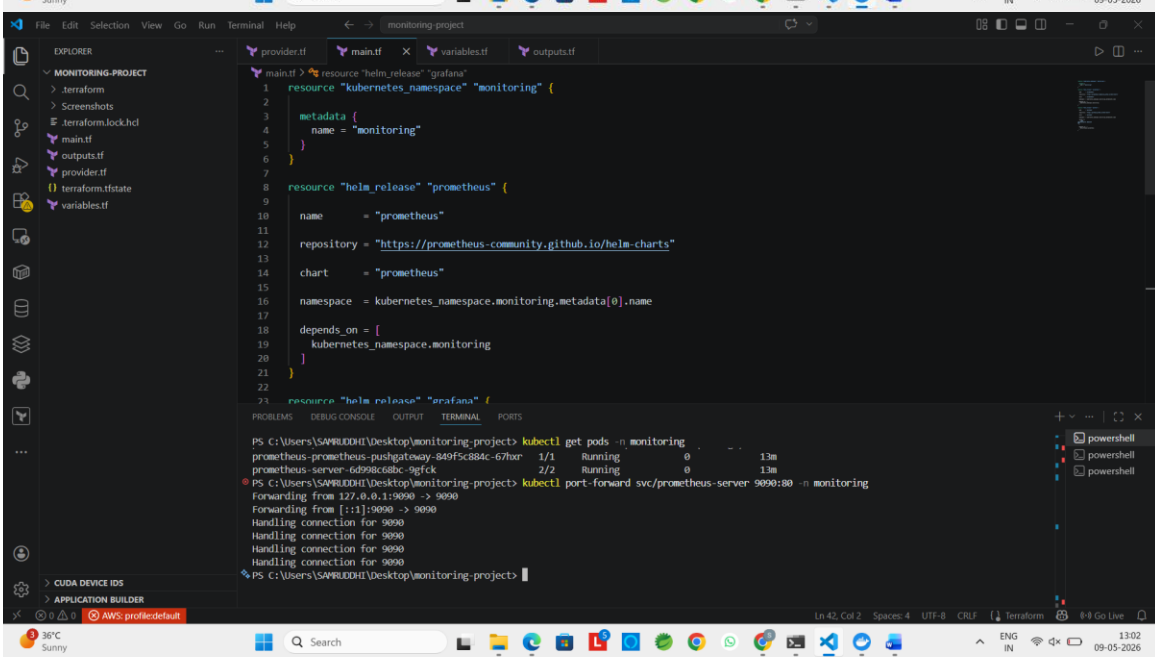

localhost:3000/connections/datasources/edit/aflr11o8ya68b
Grafana
Home › Connections › Data sources › prometheus
Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
Alerting
Connections
Add new connection
Data sources
Administration
Other
Custom query parameters
Example: max_source_resolution=5m&tin
HTTP method
POST
Series limit
40000
Use series endpoint
Exemplars
+ Add
Successfully queried the Prometheus API.
Next, you can start to visualize data by building a dashboard , or by querying data in the Explore view .
Open in Metrics Drilldown
Delete
Save & test
36°C
Sunny
12:43
09-05-2026


localhost:3000/d/rYdddlPWk/node-exporter-full?orgId=1&from=now-24h&to=now&timezone=browser&var-ds_prometheus=aflr11o8ya68b&var-job=&var-nodename=&var...
Grafana
Home › Dashboards › Node Exporter Full
Home
Bookmarks
Starred
Dashboards
Playlists
Snapshots
Library panels
Shared dashboards
Explore
Drilldown
Alerting
Connections
Add new connection
Data sources
Administration
Edit
Export
Share
Datasource
prometheus
Job
Nodename
Instance
GitHub
Grafana
Last 24 hours
Refresh
1m
Quick CPU / Mem / Disk
Pressure
No data
CPU Busy
N/A
Sys Load
N/A
RAM Us...
No data
SWAP ...
N/A
Root FS...
N/A
CPU ...
N/A
RAM ...
N/A
SWA...
N/A
Root...
N/A
Uptime
N/A
Basic CPU / Mem / Net / Disk
CPU Basic
No data
Memory Basic
No data
36°C
Sunny
12:46
09-05-2026


File Edit Selection View Go Run Terminal Help
monitoring-project
EXPLORER
MONITORING-PROJECT
.terraform
.terraform.lock.hcl
main.tf
outputs.tf
provider.tf
terraform.tfstate
variables.tf
provider.tf main.tf variables.tf outputs.tf
main.tf > resource "helm_release" "grafana"
resource "helm_release" "grafana" {
name = "grafana"
repository = "https://grafana.github.io/helm-charts"
chart = "grafana"
namespace = kubernetes_namespace.monitoring.metadata[0].name
values = [
<<EOF
adminPassword: admin123
EOF
]
depends_on = [
helm_release.prometheus
PROBLEMS DEBUG CONSOLE OUTPUT TERMINAL PORTS
PS C:\Users\SAMRUDDHI\Desktop\monitoring-project> helm list -n monitoring
NAME NAMESPACE REVISION UPDATED STATUS CHART APP VERSION
grafana monitoring 1 2026-05-09 12:25:42.9609849 +0530 IST deployed grafana-10.5.15 12.3.1
prometheus monitoring 1 2026-05-09 12:22:26.7381152 +0530 IST deployed prometheus-29.6.0 v3.11.3
PS C:\Users\SAMRUDDHI\Desktop\monitoring-project>
kubectl
kubectl
powershell
CUDA DEVICE IDS
APPLICATION BUILDER
AWS: profile:default
Ln 42, Col 2 Spaces: 4 UTF-8 CRLF Terraform Go Live
36°C Sunny
Search
ENG IN
12:47
09-05-2026


Command Prompt
error: You must be logged in to the server (the server has asked for the client to provide credentials)
C:\Users\SAMRUDDHI>minikube version
minikube version: v1.38.1
commit: c93a4cb9311efc66b90d33ea03f75f2c4120e9b0
C:\Users\SAMRUDDHI>kubectl version --client
Client Version: v1.32.2
Kustomize Version: v5.5.0
C:\Users\SAMRUDDHI>terraform --version
Terraform v1.14.3
on windows_amd64
Your version of Terraform is out of date! The latest version
is 1.15.2. You can update by downloading from https://developer.hashicorp.com/terraform/install
C:\Users\SAMRUDDHI>helm version
version.BuildInfo{Version:"v4.1.4", GitCommit:"05fa37973dc9e42b76e1d2883494c87174b6074f", GitTreeState:"clean", GoVersion:"go1.25.9", KubeClientVersion:"v1.35"}
C:\Users\SAMRUDDHI>minikube start
* minikube v1.38.1 on Microsoft Windows 11 Home Single Language 25H2
* Unable to pick a default driver. Here is what was considered, in preference order:
- hyperv: Not healthy: Hyper-V requires Administrator privileges
- hyperv: Suggestion: Right-click the PowerShell icon and select Run as Administrator to open PowerShell in elevated mode. <>
* Alternatively you could install one of these drivers:
- docker: Not installed: <nil>
- virtualbox: Not installed: unable to find VBoxManage in $PATH
- qemu2: Not installed: exec: "qemu-system-x86_64": executable file not found in %PATH%
- podman: Not installed: exec: "podman": executable file not found in %PATH%
X Exiting due to DRV_NOT_HEALTHY: Found driver(s) but none were healthy. See above for suggestions how to fix installed drivers.
C:\Users\SAMRUDDHI>kubectl get nodes
NAME STATUS ROLES AGE VERSION
docker-desktop Ready control-plane 37s v1.32.2
C:\Users\SAMRUDDHI>
36°C Sunny
Search
ENG IN
13:00
09-05-2026


docker desktop PERSONAL
Search
Ctrl+K
Settings
Give feedback
Enable Kubernetes
Start a Kubernetes single or multi-node cluster when starting Docker Desktop.
General
Resources
Docker Engine
Builders
Kubernetes
Software updates
Extensions
Features in development
Notifications
Cluster
docker-desktop
kubeadm, 1 node, v1.32.2
Running
Started 1 hour ago
Reset cluster
Cluster settings
Choose cluster provisioning method
Kubeadm
Create a singe-node cluster with kubeadm.
Version: v1.32.2
kind
SIGN IN REQUIRED
Create a cluster containing one or more nodes with kind. Requires the containerd image store
Show system containers (advanced)
Show Kubernetes internal containers when using Docker commands.
Cancel
Apply & restart
Engine running
Kubernetes running RAM 5.34 GB CPU 4.28% Disk: 5.66 GB used (limit 1006.85 GB)
Terminal
New version available
36°C
Sunny
Search
ENG
IN
13:00
09-05-2026


File Edit Selection View Go Run Terminal Help
monitoring-project
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Screenshots
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main.tf
outputs.tf
provider.tf
terraform.tfstate
variables.tf
provider.tf
main.tf
variables.tf
outputs.tf
main.tf > resource "helm_release" "grafana"
resource "kubernetes_namespace" "monitoring" {
metadata {
name = "monitoring"
}
}
resource "helm_release" "prometheus" {
name = "prometheus"
repository = "https://prometheus-community.github.io/helm-charts"
chart = "prometheus"
namespace = kubernetes_namespace.monitoring.metadata[0].name
depends_on = [
kubernetes_namespace.monitoring
]
}
resource "helm_release" "grafana" {
PROBLEMS DEBUG CONSOLE OUTPUT TERMINAL PORTS
PS C:\Users\SAMRUDDHI\Desktop\monitoring-project> kubectl get pods -n monitoring
prometheus-prometheus-pushgateway-849f5c884c-67hxr 1/1 Running 0 13m
prometheus-server-6d998c68bc-9gfck 2/2 Running 0 13m
PS C:\Users\SAMRUDDHI\Desktop\monitoring-project> kubectl port-forward svc/prometheus-server 9090:80 -n monitoring
Forwarding from 127.0.0.1:9090 -> 9090
Forwarding from [::1]:9090 -> 9090
Handling connection for 9090
Handling connection for 9090
Handling connection for 9090
Handling connection for 9090
PS C:\Users\SAMRUDDHI\Desktop\monitoring-project>
powershell
powershell
powershell
CUDA DEVICE IDS
APPLICATION BUILDER
AWS: profile:default
Ln 42, Col 2 Spaces: 4 UTF-8 CRLF Terraform Go Live
36°C
Sunny
Search
ENG
IN
13:02
09-05-2026